5.6 Composition of Functions

Name _____ Period ____ Date _____

Algebra II 5.6 · Student sheet

I can…

- evaluate a composition of functions.
- find the composition of two functions and its domain.
- use composition to solve real-life problems.

Part 1 · Coupon vs. sale

A store has a 20%-off sale, $s(x) = 0.8x$. You also have a \$10-off coupon, $c(x) = x - 10$.

Price	Sale first, then coupon	Coupon first, then sale
\$60		
\$80		
\$100		

Which order is better for the shopper? Why?

Sale first, then coupon, as one function: $c(s(x))$ = _____ Coupon first: $s(c(x))$ = _____

Part 2 · Notes: fill in the blanks

$f(g(x))$ means: find _____ first, then use that output as the _____ of f.

Work from the _____ out.

$f(g(x))$ and $g(f(x))$ are usually _____.

The domain of $f(g(x))$: x must be in the domain of _____, and $g(x)$ must be in the domain of _____.

Part 3 · Examples (work them with the class)

Example 1

Let $f(x) = 5x - 1$ and $g(x) = x^2 + 2$. Find $f(g(-1))$ and $g(f(-1))$.

Example 2

Let $f(x) = 3x - 4$ and $g(x) = x^2$. Find $f(g(x))$ and $g(f(x))$.

Example 3

Let $f(x) = 2x + 7$. Find $f(f(x))$.

Example 4

Let $f(x) = \sqrt{x}$ and $g(x) = x - 5$. Find $f(g(x))$ and its domain.

Example 5

Use the table to find $f(g(0))$ and $g(f(-1))$.

x:	-2	-1	0	1	2
$f(x)$:	3	1	-1	-3	-5
$g(x)$:	1	-2	0	2	-1

Example 6

Use the coupon $c(x) = x - 10$ and the sale $s(x) = 0.8x$. Write $c(s(x))$ and find the price of an \$85 jacket with the sale first and then the coupon.

Part 4 · Partner practice

Partner 1

Let $f(x) = x^2 - 1$ and $g(x) = 2x + 3$. Find $f(g(x))$.

Partner 2

Let $f(x) = \sqrt{x + 1}$ and $g(x) = x^3$. Find $f(g(2))$.